

Disproof of Conjecture 00000001102: the R-polynomials of S_3 do not peak at $2^d(1 - 2^{-\lceil d/2 \rceil})$

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Abstract

Conjecture 00000001102 asserts that for all $x \leq w$ in a Coxeter group, the Kazhdan–Lusztig R-polynomial satisfies $\max R_{x,w} = 2^d(1 - 2^{-\lceil d/2 \rceil})$ with $d = \ell(w) - \ell(x)$. In the symmetric group S_3 (type A_2) the R-polynomials are 1, $q - 1$, $(q - 1)^2$, $q^3 - 2q^2 + 2q - 1$ for $d = 0, 1, 2, 3$, while the conjectured values are 0, 1, 2, 6. We show that the claim fails under every reasonable reading of “ $\max R_{x,w}$ ”: the largest coefficient, the largest absolute value of a coefficient, the sum of the absolute values of the coefficients, the leading coefficient, the degree, the number of nonzero terms, the value at any fixed point q_0 , and the maximum of $q \mapsto R_{x,w}(q)$. It fails even for strict pairs $x < w$ only, and also when “max” is taken over all pairs with a given d . The recursion printed in the statement, read literally, forces $R \equiv 0$, so it cannot define anything else. Lean 4 (core library only) formalizes S_3 , its Bruhat order, the R-polynomials (characterized by their defining recursion, with existence and uniqueness proved) and all of these refutations.

1 The conjecture

The statement (English version) reads:

Definition: The R-polynomials of KL theory are defined by the recursion $R_{x,w} = qR_{xs,w} + R_{x,ws}$. *Conjecture:* For all $x \leq w$, $\max R_{x,w} = 2^{\ell(w)-\ell(x)} \cdot (1 - 2^{-\lceil (\ell(w)-\ell(x))/2 \rceil})$; that is, the peak decays as a power of two, with decay exponent the ceiling of half the length difference.

The Chinese version says the same. Write $d = \ell(w) - \ell(x)$ and

$$f(d) = 2^d(1 - 2^{-\lceil d/2 \rceil}) = 2^d - 2^{d-\lceil d/2 \rceil} = 2^d - 2^{\lfloor d/2 \rfloor},$$

so $f(0) = 0$, $f(1) = 1$, $f(2) = 2$, $f(3) = 6$, $f(4) = 12$.

Which R-polynomials. “The R-polynomials of KL theory” are the polynomials $R_{x,w} \in \mathbb{Z}[q]$ of Kazhdan and Lusztig [1], defined for a Coxeter system (W, S) by

$$T_{w^{-1}}^{-1} = \varepsilon_w q^{-\ell(w)} \sum_{x \leq w} \varepsilon_x R_{x,w}(q) T_x$$

in the Hecke algebra ($\varepsilon_x = (-1)^{\ell(x)}$). They are characterized by (see [2, §7.5], [3, Ch. 5]):

$$\begin{aligned} R_{x,w} &= 0 \text{ unless } x \leq w, & R_{w,w} &= 1, \\ \text{if } s \in S, \ ws < w : & R_{x,w} = \begin{cases} R_{xs,ws} & xs < x, \\ (q-1)R_{x,ws} + qR_{xs,ws} & xs > x. \end{cases} \end{aligned} \tag{1}$$

The recursion printed in the statement, $R_{x,w} = qR_{xs,w} + R_{x,ws}$, has no base case and no condition on s ; it is a garbled version of (1). Read literally (for all x, w and one fixed s) it forces $R \equiv 0$ (Proposition 7), which contradicts $R_{w,w} = 1$; then $\max R_{x,w} = 0 \neq 1 = f(1)$ for every covering pair anyway. We therefore use the standard R-polynomials (1).

Which group. The statement concerns KL theory in general, with no particular Coxeter group named. The symmetric group S_3 (type A_2) is a Coxeter group, so a counterexample in S_3 refutes “for all $x \leq w$ ”.

Which “max”. “ $\max R_{x,w}$ ” (the “peak”) is not defined in the statement. We consider every reading we could think of (Section 3) and refute each one, already restricted to strict pairs $x < w$. The pair $x = w$, where $R_{w,w} = 1$ but $f(0) = 0$, gives a second counterexample for every reading ρ with $\rho(1) \neq 0$.

2 S_3 , its Bruhat order and its R-polynomials

We write permutations of $\{1, 2, 3\}$ in one-line notation $w = [w(1), w(2), w(3)]$. The length $\ell(w)$ is the number of inversions. The reflections are the three transpositions; right multiplication $w \mapsto wt$ by the transposition $t = (ij)$ swaps the entries in positions i and j . The simple reflections are $s_1 = (12)$ and $s_2 = (23)$. Bruhat order is the reflexive–transitive closure of $x \rightarrow xt$ (t a reflection, $\ell(xt) > \ell(x)$). Write $e = [1, 2, 3]$, $s_1 = [2, 1, 3]$, $s_2 = [1, 3, 2]$, $s_1s_2 = [2, 3, 1]$, $s_2s_1 = [3, 1, 2]$, $w_0 = [3, 2, 1]$. The Hasse diagram has $e < s_1, s_2$, then $s_1, s_2 < s_1s_2, s_2s_1$, then $s_1s_2, s_2s_1 < w_0$, so there are 19 comparable pairs: 6 with $d = 0$, 8 with $d = 1$, 4 with $d = 2$ and one ($e < w_0$) with $d = 3$.

Proposition 1. For $x \leq w$ in S_3 , $R_{x,w}$ depends only on $d = \ell(w) - \ell(x)$:

$$R_{x,w} = 1, \quad q - 1, \quad (q - 1)^2 = q^2 - 2q + 1, \quad q^3 - 2q^2 + 2q - 1 \quad (d = 0, 1, 2, 3).$$

Proof. We apply (1). For $d = 1$, take the covering pair $e < s_1$ and $s = s_1$: $ws = e$ and $xs = s_1 > e$, so $R_{e,s_1} = (q - 1)R_{e,e} + qR_{s_1,e} = q - 1$, because $s_1 \not\leq e$. For $d = 2$, take $e < s_1s_2$ and $s = s_2$: $ws = s_1$ and $xs = s_2 > e$, so $R_{e,s_1s_2} = (q - 1)R_{e,s_1} + qR_{s_2,s_1} = (q - 1)^2$, because $s_2 \not\leq s_1$. For $d = 3$, take $w_0 = s_1s_2s_1$ and $s = s_1$: $ws = s_1s_2$ and $xs = s_1 > e$, so $R_{e,w_0} = (q - 1)R_{e,s_1s_2} + qR_{s_1,s_1s_2}$. Here $R_{s_1,s_1s_2} = (q - 1)R_{s_1,s_1} + qR_{s_1s_2,s_1} = q - 1$ (using $s = s_2$). Hence $R_{e,w_0} = (q - 1)^3 + q(q - 1) = q^3 - 2q^2 + 2q - 1$. The remaining pairs are similar. Lean checks all 36 pairs and all choices of s (Section 4), and `verify.py` recomputes everything from the Hecke algebra. It is also classical that $R_{x,w} = q - 1$ whenever $d = 1$ and $R_{x,w} = (q - 1)^2$ whenever $d = 2$, in every Coxeter group [3, Ch. 5]. $\square$

3 Refutation under every reading of “max”

Theorem 2. Let ρ be any of the following readings of “ $\max R_{x,w}$ ”: (a) the largest coefficient; (b) the largest absolute value of a coefficient; (c) the sum of the absolute values of the coefficients; (d) the leading coefficient; (e) the degree; (f) the number of nonzero terms; (g) the value $R_{x,w}(q_0)$ at a fixed point q_0 (any $q_0 \in \mathbb{C}$); (h) the maximum of the function $q \mapsto R_{x,w}(q)$ over $\mathbb{Z}$ or $\mathbb{R}$. Then the identity $\rho(R_{x,w}) = f(\ell(w) - \ell(x))$ fails for some pair $x < w$ of S_3 .

Proof. By Proposition 1, with pairs $e < s_1$ ($d = 1$), $e < s_1s_2$ ($d = 2$) and $e < w_0$ ($d = 3$):

reading	$d = 1$	$d = 2$	$d = 3$	failure
conjectured $f(d)$	1	2	6	
(a) largest coefficient	1	1	2	$d = 2: 1 \neq 2$
(b) largest coefficient	1	2	2	$d = 3: 2 \neq 6$
(c) sum of coefficients	2	4	6	$d = 1: 2 \neq 1$
(d) leading coefficient	1	1	1	$d = 2: 1 \neq 2$
(e) degree	1	2	3	$d = 3: 3 \neq 6$
(f) number of nonzero terms	2	3	4	$d = 1: 2 \neq 1$
(g) value at $q_0 = 2$	1	1	3	$d = 2: 1 \neq 2$

For (g) with an arbitrary $q_0 \in \mathbb{C}$: the case $d = 1$ requires $q_0 - 1 = 1$, so $q_0 = 2$, and then $d = 2$ gives $(q_0 - 1)^2 = 1 \neq 2$. For (h): $R_{e,s_1}(q) = q - 1$ is unbounded above on $\mathbb{Z}$ (for instance $R_{e,s_1}(3) = 2 > 1 = f(1)$), so it has no maximum, let alone the maximum $f(1) = 1$. If “max” means the maximum over a compact interval such as $[0, 1]$, then $\max_{[0,1]}(q - 1) = 0 \neq 1$. $\square$

Remark 3 (The pair $x = w$). For $x = w$ we have $d = 0$, $R_{w,w} = 1$ and $f(0) = 2^0(1 - 2^0) = 0$. So the non-strict statement “for all $x \leq w$ ” fails for every reading with $\rho(1) \neq 0$, which covers (a)–(d), (f), (g) and (h). Only the degree reading (e) survives $d = 0$, and it fails at $d = 3$.

Remark 4 (Maximum over all pairs with the same d). One might read “ $\max R_{x,w}$ ” as the maximum of $\rho(R_{x,w})$ over all pairs with the given length difference d . In S_3 all pairs with the same d have the same R-polynomial (Proposition 1), so this aggregated reading coincides with the per-pair one and fails as above. Over all Coxeter groups, $R_{x,w} = q - 1$ for $d = 1$ and $(q - 1)^2$ for $d = 2$ always, so the failures at $d \leq 2$ in the table persist. The degree is always d , so (e) fails at $d = 3$. For (b) at $d = 3$, Dyer’s formula [4] gives $\tilde{R}_{x,w}(t) = t^3 + at$ with $a \in \{0, 1\}$, hence $R_{x,w} \in \{(q - 1)^3, (q - 1)^3 + q(q - 1)\}$, whose largest absolute coefficients are 3 and 2, never 6. These general-group facts are quoted from the literature and are not formalized. `verify.py` also surveys S_4 , where the aggregated maxima are far from $f(d)$ for every reading.

Remark 5 (Other normalizations). Some authors use $\varepsilon_x \varepsilon_w R_{x,w} = (-1)^d R_{x,w}$ or the polynomial $\tilde{R}_{x,w}$ with $R_{x,w}(q) = q^{d/2} \tilde{R}_{x,w}(q^{1/2} - q^{-1/2})$. For S_3 these are $1 - q$, $(q - 1)^2$, $1 - 2q + 2q^2 - q^3$ and t , t^2 , $t^3 + t$ ($d = 1, 2, 3$). Every reading (a)–(g) still fails, at $d = 2$ or $d = 3$ (and (g) for every q_0). This is checked by `verify.py`. Readings (b), (c), (e), (f) are insensitive to the sign change and to reversing the coefficient list.

Remark 6 (Scope). The conjecture is false as a statement about KL theory, and in particular about the symmetric groups. It is not claimed false in every individual Coxeter group: in $W = \mathbb{Z}/2$ the only strict pair has $R = q - 1$, which matches $f(1) = 1$ under several readings.

Proposition 7 (The literal recursion). *Let m be an involution of a set W (e.g. $x \mapsto xs$ for a fixed simple reflection s). Let $R_{x,w}$ ($x, w \in W$) be polynomials, or even formal power series, over $\mathbb{Z}$ with $R_{x,w} = qR_{m(x),w} + R_{x,m(w)}$ for all x, w . Then $R \equiv 0$.*

Proof. Write $R_{x,w} = \sum_k r_k(x, w)q^k$. Apply the recursion to $(m(x), w)$ and $(m(x), m(w))$ and add. The terms $R_{m(x),w} + R_{m(x),m(w)}$ cancel, leaving $q(R_{x,w} + R_{x,m(w)}) = 0$. Hence $r_k(x, w) = -r_k(x, m(w))$ for all k . Now induct on k . The coefficient of q^k in the recursion is $r_k(x, w) = r_{k-1}(m(x), w) + r_k(x, m(w))$, where $r_{-1} = 0$ and $r_{k-1} = 0$ by induction. So $r_k(x, w) = r_k(x, m(w)) = -r_k(x, w)$, and $r_k(x, w) = 0$. $\square$

4 Lean formalization

The Lean project `lean4/` (Lean 4.19.0, core library only, namespace `KLR`) contains:

- **S3**: the six permutations as lists, with `S3_spec` (distinct, each a permutation of $[1, 2, 3]$). `inv` is the inversion count (length) and `swapPos/rmul` are right multiplication by transpositions and simple reflections.
- **BruhatLe**: Bruhat order, defined inductively as the reflexive–transitive closure of $x \rightarrow xt$ with $\ell(xt) > \ell(x)$. `bruhatB` is a bounded chain search, and `bruhatB_iff` proves `bruhatB x w = true` $\iff$ `BruhatLe x w` for $x \in S_3$. `bruhat_count` counts 19 comparable pairs, and `bruhat_length` shows that $x \leq w$ implies $\ell(x) \leq \ell(w)$.
- Polynomials in $\mathbb{Z}[q]$ are normalized coefficient lists. `IsRFamily R` is exactly (1): vanishing off the Bruhat order, $R_{w,w} = 1$, and the recursion for *every* simple s with $ws < w$. `R0_isRFamily` shows that the computed family satisfies it, so the recursion is consistent for every choice of descent. `RFamily_unique` shows that any two families satisfying it agree (by induction on $\ell(w)$). So all statements are about *the* R -polynomials.
- `R_classification` proves Proposition 1, and `R_values` gives $R_{e,e}, R_{e,s_1}, R_{e,s_1s_2}, R_{e,w_0}$.
- `Formula v d` is the conjectured identity multiplied by $2^{\lceil d/2 \rceil}$: $v \cdot 2^c = 2^d(2^c - 1)$ with $c = \lceil d/2 \rceil = \lfloor (d+1)/2 \rfloor$. `formula_values` shows $f(0..4) = 0, 1, 2, 6, 12$. `Claim ρ R` is “for all $x \leq w$ in S_3 , `Formula(ρ(Rx,w), d)`”. `ClaimStrict` restricts to $x \neq w$, `ClaimAgg` is the aggregated reading, and `ClaimMaxValue` is reading (h) over $\mathbb{Z}$.
- `conjecture_00000001102_false`: for every R with `IsRFamily R`, $\neg \text{Claim}$, $\neg \text{ClaimStrict}$ and $\neg \text{ClaimAgg}$ hold for readings (a)–(f) and for (g) at every integer q_0 , and $\neg \text{ClaimMaxValue}$ holds. `conjecture_00000001102_false_R0` instantiates this with the concrete family. `not_claim_diag` gives the $x = w$ counterexample for every ρ with $\rho(1) \neq 0$.
- `literal_recursion_forces_zero` is Proposition 7 for coefficient sequences $\mathbb{N} \rightarrow \mathbb{Z}$ and any involution. `literal_recursion_inconsistent` applies it to S_3 with $s = s_1$.

Not in Lean: evaluation at non-integer q_0 (the same two-line argument), the general-Coxeter-group facts of the remarks, and the alternative normalizations. `verify.py` checks the latter. There is no `sorry`, no `native_decide` and no added axioms. `#print axioms` shows at most `propext`, `Classical.choice` and `Quot.sound`.

5 Reproduction

```
cd lean4 && lake build      # Lean 4.19.0, core only
python3 verify.py          # standard library only
pdflatex report.tex
```

`verify.py` uses a different method from the Lean code. It computes $T_{w^{-1}}^{-1}$ in the Hecke algebra of S_3 (and S_4) with Laurent polynomials and reads off $R_{x,w}$. It decides Bruhat order by the tableau criterion and checks that $R_{x,w} \neq 0$ exactly when $x \leq w$. It then evaluates every reading for R , $(-1)^d R$ and $\tilde{R}$ against $f(d)$ computed with exact fractions, and searches $q_0 \in [-50, 50]$ for evaluation points.

References

- [1] D. Kazhdan and G. Lusztig, Representations of Coxeter groups and Hecke algebras, *Invent. Math.* 53 (1979), 165–184.
- [2] J. E. Humphreys, *Reflection Groups and Coxeter Groups*, Cambridge University Press, 1990.
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- [4] M. J. Dyer, Hecke algebras and shellings of Bruhat intervals, *Compositio Math.* 89 (1993), 91–115.